CME

Diagnosing Common Benign Skin Tumors

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Patients commonly present to family physicians with skin findings, and distinguishing common benign skin tumors from potentially malignant tumors is important. Benign skin tumors can often be diagnosed by their history, distribution, and characteristic morphology. A biopsy or excision is indicated if there is diagnostic uncertainty or the lesion undergoes uncharacteristic or rapid change. A keratoacanthoma is a dome-shaped nodule with a central crater and can be difficult to distinguish from squamous cell carcinomas even with dermoscopy. Pilar cysts are typically benign, but rapidly growing types could have malignant qualities. Dermoid cysts, depending on their location, can have intracranial extension if untreated. Although dermatofibromas and seborrheic keratoses are benign, atypical presentations must be differentiated from melanomas. Sebaceous hyperplasia can mimic early basal cell carcinoma. Treatment options for cherry angiomas, acrochordons, slow-growing pilar cysts, and dermatofibromas should be individualized to skin type, lesion characteristics, and the patient's cosmetic preference. Generally, excision is the treatment of choice for keratoacanthomas, rapidly proliferating pilar cysts, and dermoid cysts. Cherry angiomas are treated with laser therapy and sebaceous hyperplasia with electrodesiccation. Common treatments for acrochordons and seborrheic keratoses are shave excision and cryotherapy. Pyogenic granulomas sometimes self-involute but bleed easily and often recur at the original site. They generally respond to shave excision and electrodesiccation. In patients with darker skin, treatment with cryotherapy and laser therapy should include discussions about hypopigmentation risk.

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Family physicians often must identify and manage dermatologic diseases. A retrospective chart review over 2 years showed that about 36% of patients presenting to their primary care physician had at least one skin problem.[1] Family physicians should be familiar with characteristics of common benign skin tumors and their diagnosis, including obtaining diagnostic skin biopsies. Table 1 summarizes some of the most common benign skin lesions.[2-7]

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ACROCHORDONS

Acrochordons (skin tags) are an outgrowth of normal skin and often appear as a pedunculated lesion. They occur in nearly one-half of the general population, with increased risk between 20 and 50 years of age.[8] There is also increased prevalence among people with obesity or diabetes mellitus, likely due to hormonal imbalances. Areas of friction (e.g., axilla, neck, inguinal region) are commonly affected. Skin tags begin as small, oval excrescences that are the same color as surrounding skin and attached by a short, broad to narrow stalk.[9] They can become irritated with trauma, such as catching on jewelry or clothing (Figure 1).

The stalks are easily removed with scissor excision, cryosurgery using liquid nitrogen, or electrodesiccation if they are bothersome to the patient or for cosmetic reasons. In patients with darker skin, narrow focus of the freeze area reduces the risk of hypopigmentation.[10] Lesions are unlikely to recur after removal, but new lesions can develop in predisposed areas.

SEBACEOUS HYPERPLASIA

Sebaceous hyperplasia results from enlarged sebaceous glands. These lesions often appear as dome-shaped, asymptomatic,

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TABLE 1

Comparison of Common Benign Skin Tumors

Condition	Characteristics	Differential diagnosis	Treatment
Acrochordon (skin tag)	Pedunculated; short, broad to narrow stalk; same color as surrounding skin or darker shades of brown and black	Senescent intradermal nevus	Scissor or shave excision Cryosurgery Electrodesiccation if they are bothersome to the patient or for cosmetic reasons
Cherry angioma	Dome-shaped, small, bright red to deep purple, compressible papule	Pyogenic granuloma Amelanotic melanoma	Laser ablation Nonlaser therapy (cryotherapy, electrodesiccation, sclerotherapy)
Dermatofibroma	Solitary, firm, often hyperpigmented nodule	Dermatofibrosarcoma protuberans Kaposi sarcoma Basal cell carcinoma	No treatment required unless symptomatic or for cosmetic reasons
Dermoid cyst	Periorbital, soft, subcutaneous cyst that begins in childhood	Lipoma Epidermoid cyst	Surgical excision
Epidermoid cyst (epidermal inclusion cyst)	Firm, mobile, discrete superficial nodule with central punctum	Pilar cyst Lipoma	Intralesional steroid injection for inflamed lesion followed by surgical excision
Keloid	Elevated fibrous scar at site of previous dermal injury	Hypertrophic scaring	Intralesional injection Cryotherapy Surgical removal Immunotherapy
Keratoacanthoma	Dome-shaped nodule with a central crater	Squamous cell and basal cell carcinoma	Surgical excision
Lipoma	Soft, mobile subcutaneous mass	Liposarcoma Epidermoid or pilar cyst	Incision with manual expression
Pilar cyst	Single or multiple soft, subcutaneous nodules	Lipoma Epidermoid cyst	Generally treated for cosmesis Surgical excision if rapidly growing or proliferating
Plantar wart	Single or multiple, sandpaper-like lesions on plantar surface of the foot, may have black pinpoints	Heloma (corn)	Salicylic acid Cryosurgery Electrodesiccation Immunotherapy

Clinical considerations
Associated with obesity and diabetes mellitus
Increased number associated with Fabry disease
Associated with the dimple sign (Fitzpatrick sign) Multiple lesions may be associated with an underlying systemic immune disorder (e.g., systemic lupus erythematosus, HIV)
Craniofacial origin; most are uncomplicated, but they may become complex and extend, requiring surgical coordination
Recurrent lesions can become more difficult to remove
There is a 15-fold increase in risk of keloids in people with dark skin tones
Difficult to differentiate from squamous cell carcinoma on examination and dermoscopy; excision should be performed to diagnose and treat accordingly
Recurrence is common after removal
Likely benign, especially when soft; rare malignant potential
Can spread through direct contact

continues ➤

yellow papules with central umbilication, primarily on the forehead and cheeks. Hair follicles surrounded by yellow lobules on dermoscopy are diagnostic, and there can be single or multiple 1- to 4-mm lesions (Figure 2). No treatment is required. However, if desired for cosmetic reasons, therapeutic options include cryosurgery, shave excision, laser ablation, electrodesiccation with curettage, and chemical cautery. Oral isotretinoin can be used for widespread lesions.[11,12]

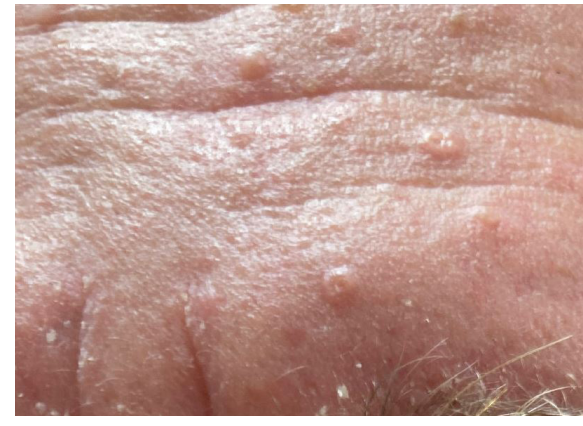
Sebaceous hyperplasia is benign but should be differentiated from basal cell carcinoma, which can be similar in appearance and location. Basal cell carcinoma generally appears more pink than yellow and enlarges over time. Dermatoscopic visualization of sebaceous hyperplasia reveals crown vessels at the base of the follicle and between lobules, whereas arborizing vessels and shiny white structures are seen with basal cell carcinoma.[13] If lesions change in color or size, shave removal can be performed to rule out basal cell carcinoma.[14,15]

FIGURE 1



Acrochordon (skin tag) on the leg, with inflammatory changes. Histology included partial infarction and reactive and reparative changes.

FIGURE 2



Sebaceous hyperplasia on the forehead.

KERATOACANTHOMAS

Keratoacanthoma is an epithelial tumor that presents as a nodule with a central crater. The annual incidence is 104 out of 100,000 people, with peak occurrence between 65 and 71 years of age.[16] It typically begins as a smooth, dome-shaped, red papule (resembling molluscum contagiosum) that expands over weeks to a 1- to 2-cm lesion with a central keratin-filled crater (Figure 3). Sun-exposed skin, such as arms and hands, is the most common location, followed by the trunk. Dermoscopy findings of keratoacanthomas share some features with squamous cell carcinoma and cannot be used to differentiate the two conditions.[17]

If untreated, keratoacanthomas usually stop growing in 6 weeks and regress slowly over 2 to 12 months, although some may progress. First-line therapy is typically surgical excision and histopathologic evaluation to confirm the diagnosis and completeness of tumor removal.[17,18] Due to the difficulty in distinguishing keratoacanthomas from cutaneous squamous

TABLE 1 (continued)

Comparison of Common Benign Skin Tumors

Condition	Characteristics	Differential diagnosis	Treatment
Pyogenic granuloma	Red, yellow, or purple, friable papule or nodule; often surrounded by a scaly red or brown collarette; grows rapidly before stabilizing; bleeds easily	Spitz nevus Basal cell carcinoma Squamous cell carcinoma Amelanotic melanoma	Laser ablation Shave excision with electrodesiccation at the base Topical beta blockers are effective in children but used only with definitive treatment in adults
Sebaceous hyperplasia	Dome-shaped, yellow papules around hair follicles with central umbilication; yellow lobules on dermoscopy	Basal cell carcinoma	No treatment required If treatment desired for cosmetic reasons, options include cryosurgery, shave excision, laser ablation, electrodesiccation with curettage, chemical cautery, and oral isotretinoin
Seborrheic keratoses	Well-circumscribed, raised macules, papules, or plaques with a "stuck-on" appearance; same color as surrounding skin or darker	Atypical nevus Melanoma	No treatment required If treatment desired, options include shave excision, cryosurgery, laser ablation, topical hydrogen peroxide

Information from references 2-7.

FIGURE 3



Keratoacanthoma, which often mimics squamous cell carcinoma.

cell carcinoma, lesions suspicious for keratoacanthoma should be excised with a 5-mm margin.[17,18] Mohs surgery, which is tissue sparing, can be considered for tumors that are deep or arise on the central face, ear, nose, and periocular or perioral skin. Other treatment options after obtaining a skin biopsy specimen include electrodesiccation and curettage, intralesional therapy with fluorouracil or methotrexate, radiation, and topical therapy. Early treatment may improve cosmetic outcomes by limiting damage to the skin and underlying structures.[17]

Clinical considerations
Rule out melanoma
Refer patients with facial lesions
Lesions concerning for basal cell carcinoma should be removed and submitted for histopathologic evaluation
Malignancy workup should be considered if multiple lesions develop rapidly

PYOGENIC GRANULOMAS

Pyogenic granulomas are rapidly growing papules that develop over weeks to months before stabilizing at less than 2 cm in size. Lesions are more common on the head and neck in children and on the trunk and extremities in adults. They are also common during pregnancy. Pyogenic granulomas are friable papules or nodules that can be yellow, red, or purple (Figure 4). These vascular tumors are often surrounded by a scaly red, brown, or white collarette (Figure 5). They often bleed or ulcerate, are difficult to control, and recur at the original site (unlike cherry angiomas). They sometimes regress spontaneously, but most patients receive treatment. The differential diagnosis includes cherry angiomas, Spitz nevi, amelanotic melanoma, and squamous or basal cell carcinoma.

A wide variety of treatment modalities have been studied, most commonly surgical excision, which offers the lowest overall recurrence rates and is most effective for larger lesions. Other effective treatment modalities include cryotherapy, electrodesiccation, and laser therapy.[19,20] Studies in children show that topical beta blockers (e.g., timolol, propranolol) with occlusion are effective in treating pyogenic granuloma after several weeks. However, only a few small studies have evaluated the effectiveness of beta blockers in adults, and use was limited to adjunctive therapy with definitive treatment options.[20,21] Recurrence rates for pyogenic granulomas range from 0% to 15% (average 4.5%) in small treatment studies.[19]

DERMATOFIBROMAS

Dermatofibromas (benign fibrous histiocytoma) originate from a dermal proliferation of fibroblasts. These lesions can occur after minor trauma, insect bites, viral infections, ruptured cysts, or folliculitis. They typically present as a solitary, firm, often hyperpigmented, 3- to 10-mm nodule (Figure 6). Lesions are often asymptomatic but can be pruritic when irritated. Diagnosis is usually based on clinical appearance and history. The lesions display dimple sign (Fitzpatrick sign), in

FIGURE 4



Pyogenic granuloma on the digit presenting after trauma.

FIGURE 5



Pyogenic granuloma on the palm, with scaly collarette.

which lateral pressure on the lesion produces a dimpling or retraction beneath the skin[2,22] (Figure 7).

Although a solitary dermatofibroma may be an incidental finding, multiple dermatofibromas may be associated with an underlying systemic immune disorder, such as systemic lupus erythematosus or HIV.[23] No treatment is required unless the lesion is symptomatic or the patient requests excision for cosmetic reasons. Excisional biopsy is preferred over shave biopsy to ensure clear histology and complete removal of the lesion, especially with atypical variants or suspicion for malignancy (e.g., lesion is painful, changes in shape or character, or is persistently pruritic).

SEBORRHEIC KERATOSES

Seborrheic keratoses, common after 30 years of age, are comprised of epidermal keratinocytes. They present as well-circumscribed, raised lesions with a "stuck on" appearance and color ranging from that of the surrounding skin or darker. Seborrheic keratoses can present as macules, papules, or plaques (Figure 8). The back is the most common location, but they can also occur on the abdomen, extremities, and face. Although usually asymptomatic, pruritus around the lesion is the most common symptom. Seborrheic keratoses do not require removal but are easily treated cosmetically with shave biopsy, cryosurgery, laser ablation, and topical hydrogen peroxide.[24,25] Treatment of seborrheic keratoses with cryotherapy should involve counseling, as it can lead to hypopigmentation.[10]

Although rare, extensive seborrheic keratoses should alert the clinician to the possibility of an underlying systemic malignancy because seborrheic keratoses tend to proliferate with malignancy and regress with treatment.[2,26,27] Due to their dark coloration, differentiating seborrheic keratoses from melanoma is important. Melanomas tend to vary more in color, with browns, blues, black, and grays.[28] Dermatoscopic evaluation can help to differentiate these lesions. Dermatosis papulosa nigra lesions are histologically similar to seborrheic keratoses but have photodistribution on the face, head, and neck and occur in patients with darker skin tones (Figure 9).

CHERRY ANGIOMAS

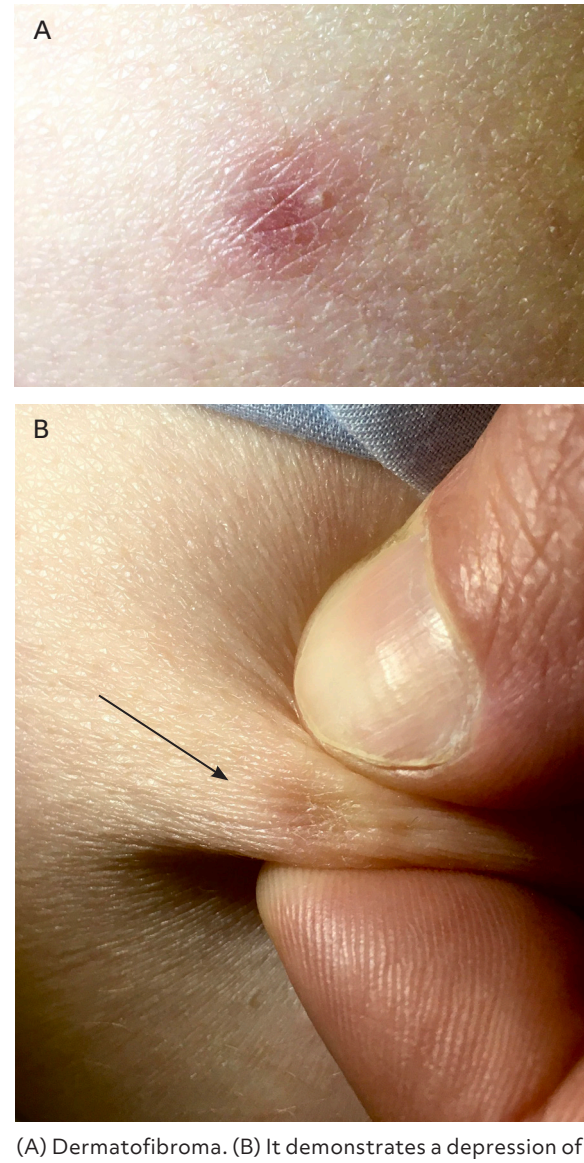
Cherry angiomas, also known as De Morgan spots, are vascular papules that appear with increasing age. They are related to, but distinct from, infantile hemangiomas. They are dome-shaped, small (1 to 5 mm), bright red to deep purple, soft,

FIGURE 6.



Dermatofibromas on the arm.

FIGURE 7



(A) Dermatofibroma. (B) It demonstrates a depression of the entire lesion (secondary to the dermal scarring seen as the major pathologic change) on lateral compression of the lesion with the thumb and index finger, also known as the Fitzpatrick or dimple sign (arrow).

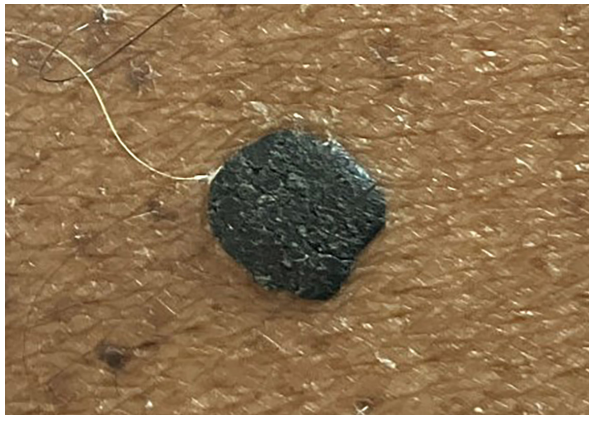
compressible papules that bleed profusely with traumatic rupture and may blanch with pressure (Figure 10). They occur primarily on the trunk but also appear on the extremities. A recent review showed that laser therapy and nonlaser therapy (cryotherapy, electrodesiccation, sclerotherapy) are equally effective for cosmesis.[29]

Although cherry angiomas do not have malignant potential, in small studies, high numbers of cherry angiomas were associated with lighter skin color and a history of melanoma.[30] The differential diagnosis includes pyogenic granuloma and amelanotic melanoma, which grows or changes over time and is often asymmetrical.

PILAR CYSTS

Pilar cysts (trichilemmal cysts) are slow-growing, intradermal cysts derived from outer hair follicle root sheath tissue and are found primarily on the scalp (Figure 11). These cysts have an appearance similar to epidermoid cysts and are sporadic or hereditary in patients with multiple pilar cysts.[31] On ultrasonography, pilar cysts have increased echogenicity and more cystic features than epidermoid cysts.[4] Although most pilar cysts are benign and slow-growing, rare forms, referred to as proliferating trichilemmal tumors, proliferate, grow rapidly, and have overlying skin changes and inflammation. This transitional type warrants excision because of the potential for malignant

FIGURE 8



Seborrheic keratosis on the arm.

FIGURE 9



Dermatosis papulosa nigra, a variant of seborrheic keratosis, with photodistribution on the face and neck.

FIGURE 10



Multiple, slightly raised cherry angiomas on the torso.

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FIGURE 11



Soft, mobile pilar cyst present on the scalp for years.

SORT: KEY RECOMMENDATIONS FOR PRACTICE

Clinical recommendation	Evidence rating	Comments
Due to the difficulty in distinguishing keratoacanthomas from cutaneous squamous cell carcinoma, lesions suspicious for keratoacanthoma should be excised with a 5-mm margin.[17,18]	C	Observational study and expert consensus
Although treatment of pyogenic granuloma is predominantly surgical, topical beta blockers are effective in children.[20,21]	B	Multiple small clinical studies
Seborrheic keratoses are common benign lesions that may be treated cosmetically. However, a sudden increase in seborrheic keratosis lesions can indicate an underlying malignancy.[2,13]	C	Expert consensus
Suspected dermoid cysts in infants and children should be referred for excision because they have the potential for intracranial extension.[5,36]	C	Expert consensus

A = consistent, good-quality patient-oriented evidence; B = inconsistent or limited-quality patient-oriented evidence; C = consensus, disease-oriented evidence, usual practice, expert opinion, or case series. For information about the SORT evidence rating system, go to https://www.aafp.org/afpsort.

transformation (malignant trichilemmal cyst), accounting for less than 0.1% of all skin cancers.[32] Rapidly growing pilar cysts are definitively diagnosed and treated with complete surgical excision and monitoring for recurrence.

EPIDERMOID CYSTS

Epidermoid cysts (epidermal inclusion cysts) are the most common type of cutaneous cyst. They are often called sebaceous cysts, but they do not contain sebaceous cells or material. The epidermoid cyst wall is stratified squamous epithelium that proliferates and results in an accumulation of keratin material within the dermis.[33] Diagnosis is based on appearance and palpation of a firm, mobile, discrete superficial cyst or nodule. Careful inspection often reveals a central punctum. Epidermoid cysts usually present on the face, neck, or trunk. They can remain stable for prolonged periods or become inflamed, become infected, or spontaneously rupture.

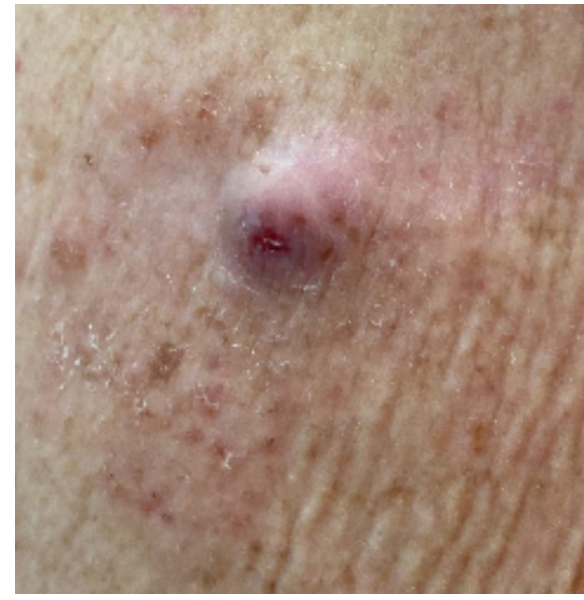
Treatment is unnecessary unless desired by the patient and can be accomplished via simple excision with removal of the cyst and cyst wall. Inflamed or ruptured cysts often resolve spontaneously without therapy, although they may recur. Intralesional steroid injections can hasten resolution of inflamed cysts, followed by interval excision.[34]

The presence of multiple epidermoid cysts (Figure 12), or cysts in atypical locations (e.g., extremities), is associated with the autosomal dominant Gardner syndrome, colon polyps, and colon cancer. Epidermoid cysts are benign; however, malignancies develop rarely within or adjacent to the cyst (e.g., basal cell carcinoma, Bowen disease, squamous cell carcinoma, mycosis fungoides, melanoma in situ).[2,35]

DERMOID CYSTS

Although dermoid cysts are uncommon overall, they occur more often in infants and children. If untreated, small dermoid cysts grow and persist into adulthood with the potential for intracranial extension.[36] Dermoid cysts are subcutaneous and have embryological origins associated with craniofacial development. They contain a mixture of squamous epithelium, keratin, sebaceous glands, and hair.[6] Dermoid cysts occur primarily in the periorbital region, but can occur elsewhere on the head and neck, and are classified by their location[5] (Figure 13). Imaging of the orbits can help differentiate superficial, freely mobile cysts

FIGURE 12



Multiple epidermoid (epidermal inclusion) cysts, with the characteristic central punctum, on the arm of a patient with colon polyps.

FIGURE 13



Dermoid cyst that developed in childhood.

from deeper cysts fixed to the underlying bone and may require coordination with a surgical specialist.[5,36] Due to risk of deeper involvement, patients with a suspected dermoid cyst should be referred to an ear, nose, and throat specialist; dermatologist; or oculoplastic surgeon, depending on the location of the cyst.[5,36]

This article updates previous articles on this topic by Higgins, et al.,[2] and Luba, et al.[28]

Data Sources: PubMed searches were completed in Clinical Queries using the key terms acrochordon, sebaceous hyperplasia, keratoacanthoma, pyogenic granuloma, dermatofibroma, seborrheic keratosis, cherry angiomas, dermoid cysts, epidermoid cyst, and pilar cysts. Other key words searched in PubMed: benign skin lesions, benign skin tumors, skin diseases, diagnosis, and treatment. The search included reviews, meta-analyses, randomized controlled trials, and clinical trials. We also searched Essential Evidence Plus, the Cochrane database, and UpToDate. We used references from 2006 and 2003 *AFP* articles on benign skin tumors. We critically reviewed studies that used patient categories such as race and/or gender but did not define how these categories were assigned, stating their limitations in the text. Search dates: June 14, 2023, and August 28, 2024.

REFERENCES

1. Lowell BA, et al. Dermatology in primary care. *J Am Acad Dermatol*. 2001;45(2):250-255.
2. Higgins JC, et al. Diagnosing common benign skin tumors. *Am Fam Physician*. 2015;92(7):601-607.
3. Juckett G, et al. Management of keloids and hypertrophic scars. *Am Fam Physician*. 2009;80(3):253-260.
4. Ren WW, Wu L, Wang Q, et al. The value of ultrasound for differentiating trichilemmal cysts from epidermoid cysts. *J Ultrasound Med*. 2023;42(9):1941-1950.
5. Golden BA, Jaskolka MS, Ruiz RL. Craniofacial and orbital dermoids in children. *Oral Maxillofac Surg Clin North Am*. 2012;24(3):417-425.
6. Reissis D, Pfaff MJ, Patel A, et al. Craniofacial dermoid cysts. *Yale J Biol Med*. 2014;87(3):349-357.
7. American Academy of Dermatology Association. Warts: overview. https://www.aad.org/public/diseases/a-z/warts-overview
8. Banik R, Lubach D. Skin tags. *Dermatologica*. 1987;174(4):180-183.
9. Belgam Syed SY, Lipoff JB, Chatterjee K. Acrochordon. *StatPearls*. August 8, 2023. Accessed August 8, 2023. https://www.ncbi.nlm.nih.gov/books/NBK448169/
10. Clebak KT, Mendez-Miller M, Croad J. Cutaneous cryosurgery for common skin conditions. *Am Fam Physician*. 2020;101(7):399-406.
11. Liu A, Taylor MB, Sotoodian B. Treatment of sebaceous hyperplasia by laser modalities. *J Drugs Dermatol*. 2020;19(5):547-552.
12. Yu C, Shahsavari M, Stevens G, et al. Isotretinoin as monotherapy for sebaceous hyperplasia. *J Drugs Dermatol*. 2010;9(6):699-701.
13. Marghoob AA, Usatine RP, Jaimes N. Dermoscopy for the family physician. *Am Fam Physician*. 2013;88(7):441-450.
14. *Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology*. 9th ed. McGraw-Hill Education; 2023.
15. Dermnet. Sebaceous hyperplasia. Updated June 2014. Accessed June 2023. https://dermnetnz.org/topics/sebaceous-hyperplasia
16. Sullivan JJ. Keratoacanthoma: the Australian experience. *Australas J Dermatol*. 1997;38(suppl 1):S36-S39.
17. Kwiek B, Schwartz RA. Keratoacanthoma (KA): an update and review. *J Am Acad Dermatol*. 2016;74(6):1220-1233.
18. Tran DC, Li S, Henry S, et al. An 18-year retrospective study on the outcomes of keratoacanthomas with different treatment modalities at a single academic centre. *Br J Dermatol*. 2017;177(6):1749-1751.
19. Lee J, Sinno H, Tahiri Y, et al. Treatment options for cutaneous pyogenic granulomas. *J Plast Reconstr Aesthet Surg*. 2011;64(9):1216-1220.
20. Plachouri KM, Georgiou S. Therapeutic approaches to pyogenic granuloma: an updated review. *Int J Dermatol*. 2019;58(6):642-648.
21. Dany M. Beta-blockers for pyogenic granuloma. *J Drugs Dermatol*. 2019;18(10):1006-1010.
22. Zaballos P, Puig S, Llambrich A, et al. Dermoscopy of dermatofibromas. *Arch Dermatol*. 2008;144(1):75-83.
23. Beatrous SV, et al. Associated conditions in patients with multiple dermatofibromas. *Dermatol Online J*. 2017;23(9):13030/qt8zv852d8.
24. Ahlgrimm-Siess V, Cao T, Oliviero M, et al. Seborrheic keratosis. *J Am Acad Dermatol*. 2013;69(1):120-126.
25. Baumann LS, Blauvelt A, Draelos ZD, et al. Safety and efficacy of hydrogen peroxide topical solution, 40% (w/w), in patients with seborrheic keratoses. *J Am Acad Dermatol*. 2018;79(5):869-877.
26. Bernett CN, Davidson CL, Schmieder GJ. Leser Trélat sign. StatPearls. Updated February 2024. Accessed March 1, 2024. https://www.ncbi.nlm.nih.gov/books/NBK470554/
27. Wollina U. Recent advances in managing and understanding seborrheic keratosis. F1000Res. 2019;8:F1000.
28. Luba MC, Bangs SA, Mohler AM, et al. Common benign skin tumors. *Am Fam Physician*. 2003;67(4):729-738.
29. Buslach N, Foulad DP, Saedi N, et al. Treatment modalities for cherry angiomas: a systematic review. *Dermatol Surg*. 2020;46(12):1691-1697.
30. Betz-Stablein B, et al. Anatomic distribution of cherry angiomas in the general population. *Dermatology*. 2022;238(1):18-26.
31. Hörer S, et al. A monoallelic two-hit mechanism in plcd1 explains the genetic pathogenesis of hereditary trichilemmal cyst formation. *J Invest Dermatol*. 2019;139(10):2154-2163.e5.
32. Lee DY, et al. A malignant proliferating trichilemmal cyst arising on the elbow of a man. *Medicine (Baltimore)*. 2023;102(25):e34035.
33. Hoang VT, Trinh CT, Nguyen CH, et al. Overview of epidermoid cyst. *Eur J Radiol Open*. 2019;6:291-301.
34. Zuber TJ. Minimal excision technique for epidermoid (sebaceous) cysts. *Am Fam Physician*. 2002;65(7):1409-1412, 1417-1418, 1420.
35. Swygert KE, Parrish CA, Cashman RE, et al. Melanoma in situ involving an epidermal inclusion (infundibular) cyst. *Am J Dermatopathol*. 2007;29(6):564-565.
36. Pryor SG, Lewis JE, Weaver AL, et al. Pediatric dermoid cysts of the head and neck. *Otolaryngol Head Neck Surg*. 2005;132(6):938-942.

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